EXATHON INDIA 2026 · CIVIC / SUSTAINABILITY

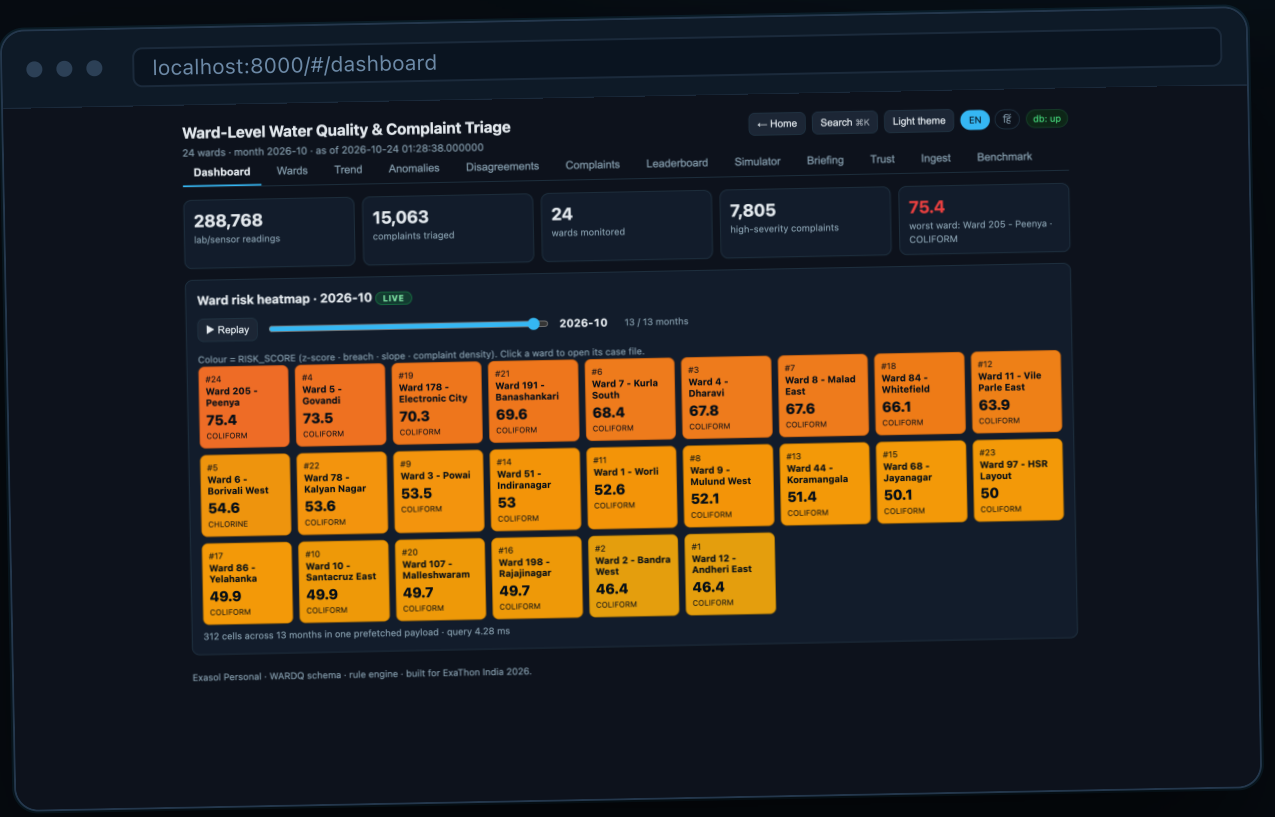
WARDQ

Catch the water failing — before the clinic fills.

Ward-level water quality and complaint triage. One page each morning: every ward ranked, every breach explained, every number traceable to the SQL that produced it.

1,002,700 residents · **24** wards · **288,768** readings · **213** tests

runs offline · ₹0 forever



BUILT SOLO IN 48 HOURS · EXASOL PERSONAL · 2 VCPU / 6 GB

Three people need the same answer, at three different speeds.

One dataset, three jobs, and no shared surface between them. The gap is not missing data — it is missing timing.

WHO · THE WARD HEALTH OFFICER

Ranks complaints by noise

Holds the citizen signal but not the instruments, so the morning list is ordered by who complained loudest rather than by which ward is actually deteriorating.

WHO · THE UTILITY ENGINEER

Cannot tell a bad probe from bad water

Probe alarms and lab confirmations sit in two registers that never meet. One ward shows **22 pH alarms** and **zero** confirmations — so the safe habit becomes ignoring alarms.

WHO · 1,002,700 RESIDENTS

Finds out last

The wards downstream of the intake pay for contamination they did not produce, and their first real warning is the clinic queue.

WHY IT MATTERS

By the time illness confirms a water failure, the exposure window has closed: the only action left is *response*. Before it, a flushing order, a chlorine correction or a probe calibration is still *prevention* — and a municipal dashboard that reports the week the illness is counted cannot tell the two apart.

9 of 24	**24** of 96	**26** days	**0**
WARDS DOWNSTREAM OF INTAKE	PAIRS OVER LIMIT THIS MONTH	NOTICE BEFORE THE NEXT CROSSING	OF THEM IN AN ILLNESS REPORT YET

WHY IT HAPPENS · THREE CLOCKS

Water fails first. The evidence arrives weeks later.

A municipal water supply gives three signals on three completely different clocks. Every dashboard in the building reports the last one — the only one that is already too late.

CLOCK 1 · HOURS

The probe moves

Turbidity doubles at 11 pm and residual chlorine decays below 0.2 mg/L. The instrument knows the same night.

CLOCK 2 · 7–14 DAYS

The lab confirms

Coliform plates need incubation and a technician. Confirmation is a report, not an alarm.

CLOCK 3 · WEEKS

The clinic fills

Diarrhoeal cases show up in PHC registers after the exposure window has already closed.

instrument knowledge — the 3-week window where an intervention still changes the outcome — human cost

THE WINDOW THIS PRODUCT EXISTS FOR

Between the probe firing and the clinic filling there are about three weeks in which an intervention still changes the outcome. Nothing in the municipal stack is built to act inside that window — and that window is the product.

WARDQ does not wait for clock three. It reports clock one, prices clock two, and turns “this ward is getting worse” into a dated work order:

```
Ward 191 · Turbidity 4.695 NTU · +0.0118/day · crosses 5 NTU ≈ 17 Nov
```

THE PRODUCT

One page, every morning: who is worst, why, and what to do.

Not another map. A triage sheet a ward officer can act on before 9 am — with the two things missing from every civic dashboard: a dated forecast, and a receipt for every number.

01 · RANK

One score, four measured parts

`RISK_SCORE = 0.35·|z| + 0.30·breach days + 0.15·slope + 0.20·complaint density`

Computed in-database by a Lua UDF over 288,768 rows, materialised so the page never waits on a window function.

02 · EXPLAIN

"Why" is a link, not a paragraph

Every figure carries a `why this · file` chip that opens the exact SQL in place. Eleven query recipes, one lineage registry, zero hand-typed numbers.

03 · ACT

The brief leaves the app

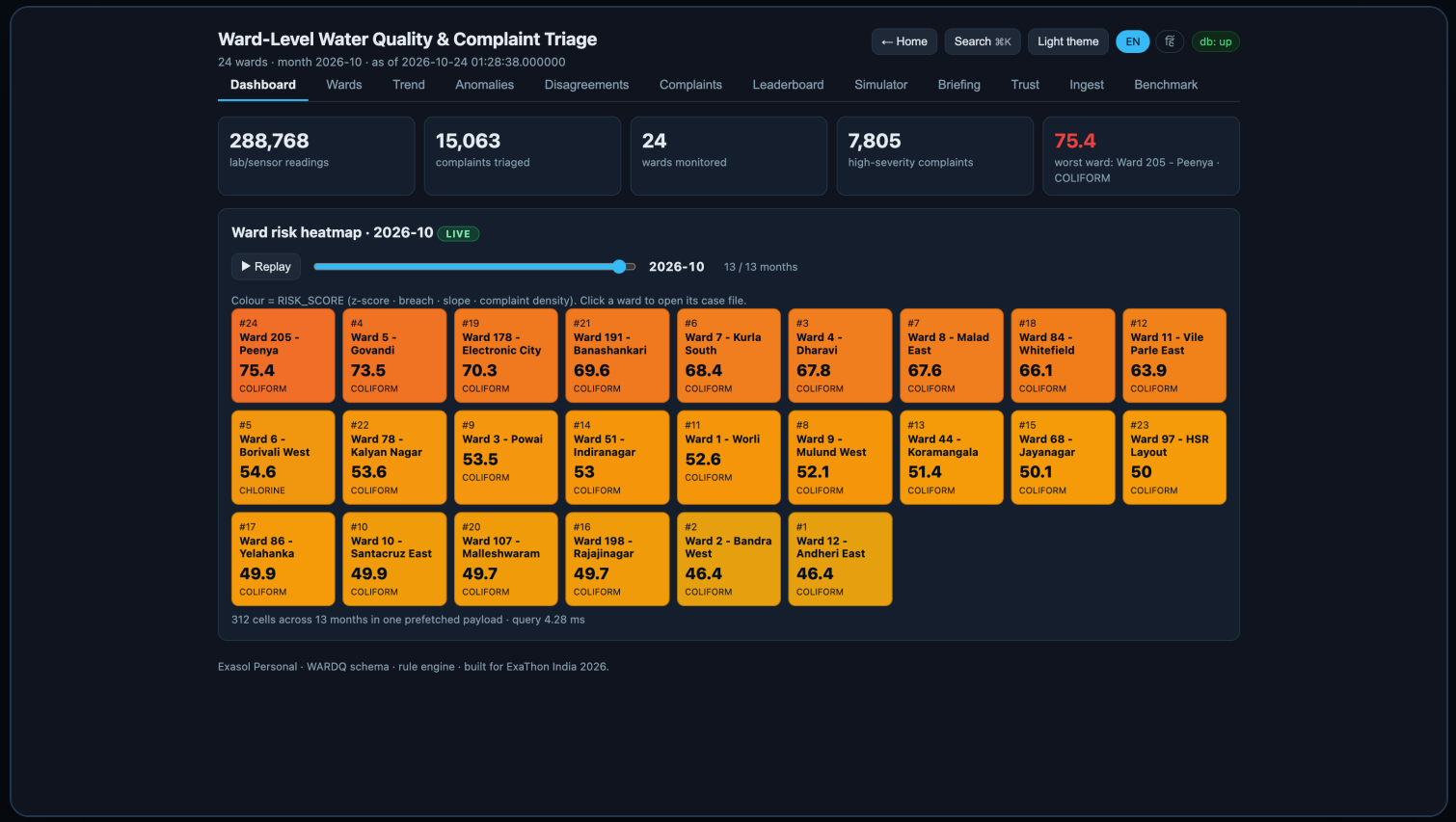
Three ranked actions, downloadable as Markdown and CSV; each 90-day breach ETA also lands in the officer's calendar as a `.ics` with a 7-day alarm, plus a pull-based Atom feed.

THE RECEIPT

Every tile opens `why this · db/sql/…` — the SQL file, the window, the row count and the timestamp that produced the number on screen. A judge can check any figure in two clicks.

288,768	15,063	96	4.3ms	<1s
LAB + SENSOR READINGS	COMPLAINTS TRIAGED	WARD × PARAM FORECASTS	13-MONTH HEATMAP PAYLOAD	DASHBOARD LOAD, LOCAL

Every ward, ranked by risk — replayable across 13 months.



Live capture · Exasol Personal · month 2026-10 · 24 wards

- **Colour is the score**, not a guess: orange to amber over four measured components.
- **Replay** scrubs the whole 24 × 13 grid — the same prefetched payload, 4.28 ms.
- **Worst ward first**: Ward 205 Peenya at 75.4, worst parameter coliform, 10,050 connections.
- **Every tile is a door** — click through to the case file without losing the month.

BEFORE A SINGLE COMPLAINT

24 of 96 ward × parameter pairs are already over their limit in the current month. The ranking is built from the instruments, then corroborated by the citizens — never the other way round.

A score you can take apart — bar by bar.



Ward 24 · Ward 205 – Peenya · Bengaluru North · 43,900 people · 10,050 connections · downstream

- **Four bars, four weights** — anomaly |z|, days in breach, 12-month slope, complaint density — that multiply out to 75.4.
- **13-month risk ribbon** shows whether this is a bad month or a bad trend.
- **Breach forecast per parameter**, each with a confidence label and the ±1.96σ band at 90 days.
- **Print report** and a deep link (`#/ward/24`) that survives a reload.

WHY THE TRIBUNAL IS SHORT

The weights live in the UDF, not in a slide deck. Change them, re-materialise, and every ward on every screen moves together — an auditable policy, not a tuned opinion.

A dated breach forecast, with an honest error bar.

Two years of daily readings become 108 weekly buckets, an ordinary-least-squares slope over trailing 12 months, and a 90-day projection band that widens as √(days/7) — because a straight line deserves to look less certain the further it is dragged.



Ward 11 · Vile Parle East · turbidity, monsoon intrusions visible as two cliff edges · 108 buckets · 6.97 ms

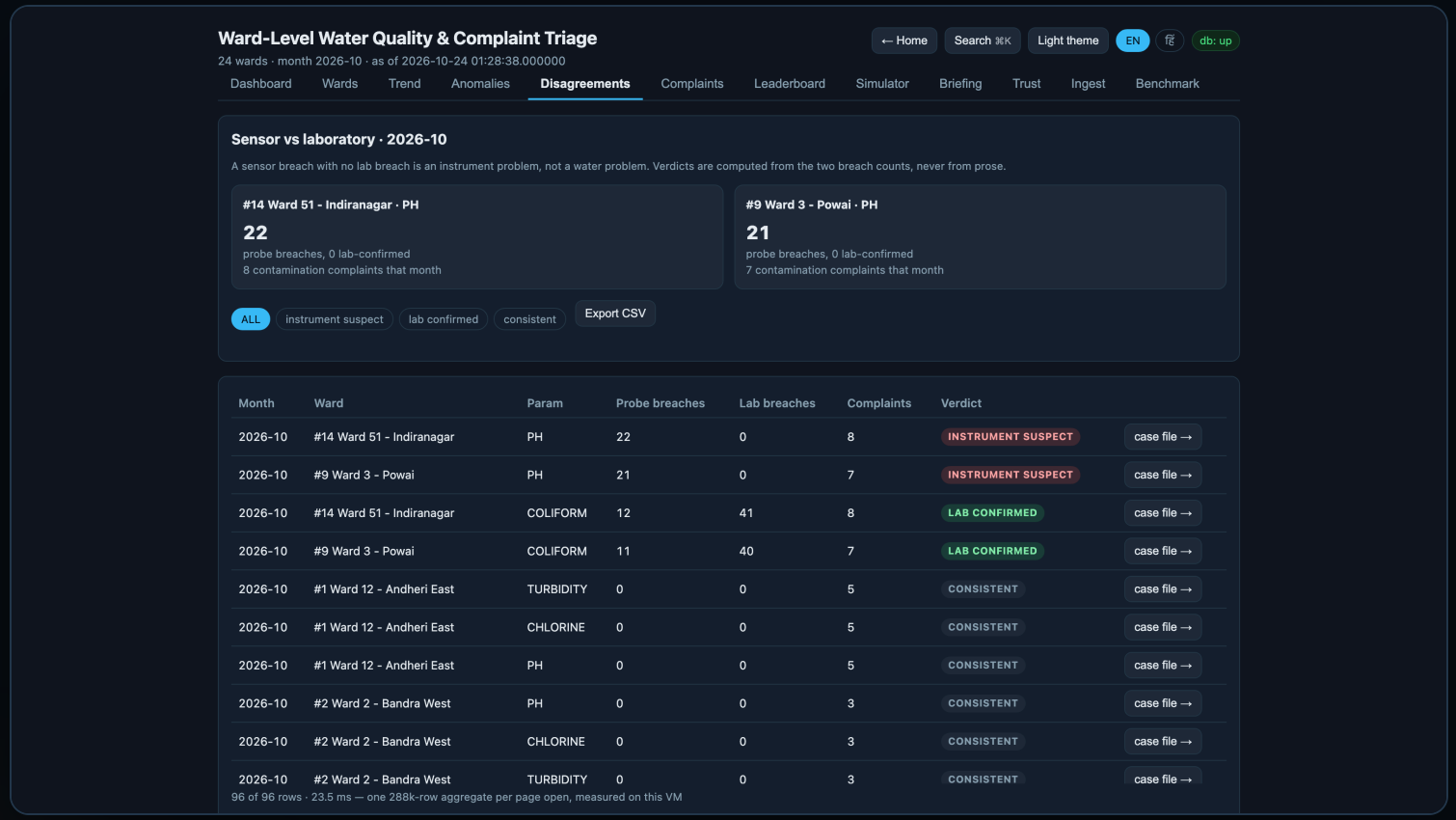
THE LINE THAT MATTERS

4.695 NTU today, **+0.0118/day** over 12 months → crosses the **5 NTU** limit in **26 days**.

Ward 191 · Banashankari. High confidence, 31 samples. That is a flushing order three weeks before a boil notice.

- **Every parameter** gets its own verdict: stable, over limit, or a dated crossing.
- **Status is computed** from slope and threshold — ≥`90d`, ~`26d` or `over limit`.
- **The same number** drives the calendar alarm and the morning brief.

A sensor breach with no lab breach is an instrument problem, not a water problem.



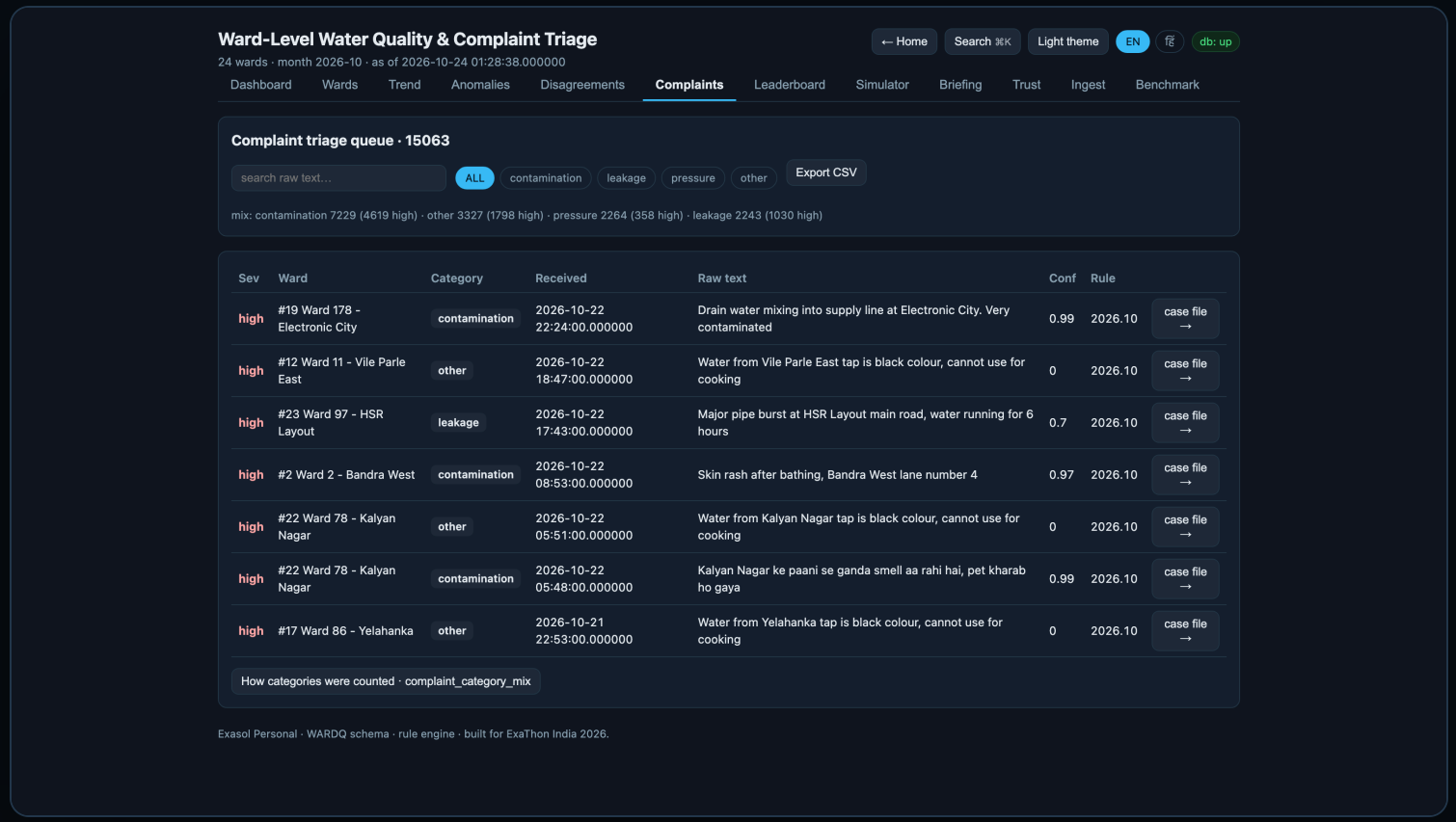
96 of 96 ward × parameter rows · 23.5 ms over a 288k-row aggregate

- **Three verdicts, computed** from the two breach counts — never from prose: instrument suspect, lab confirmed, consistent.
- **Ward 51 · Indiranagar**: 22 pH probe breaches, **0** lab-confirmed, 8 complaints → send a technician, not a tanker.
- **Ward 3 · Powai**: 11 coliform probe breaches and 40 lab-confirmed → this one is real.
- **Filter and export** the whole table; judges walk away with the CSV.

WHY THIS WINS ROOMS

Most teams only know how to raise alarms. This screen knows how to *withdraw* one — and that is the difference between a dashboard and a dispatch desk.

15,063 WhatsApp complaints, triaged by 169 patterns.



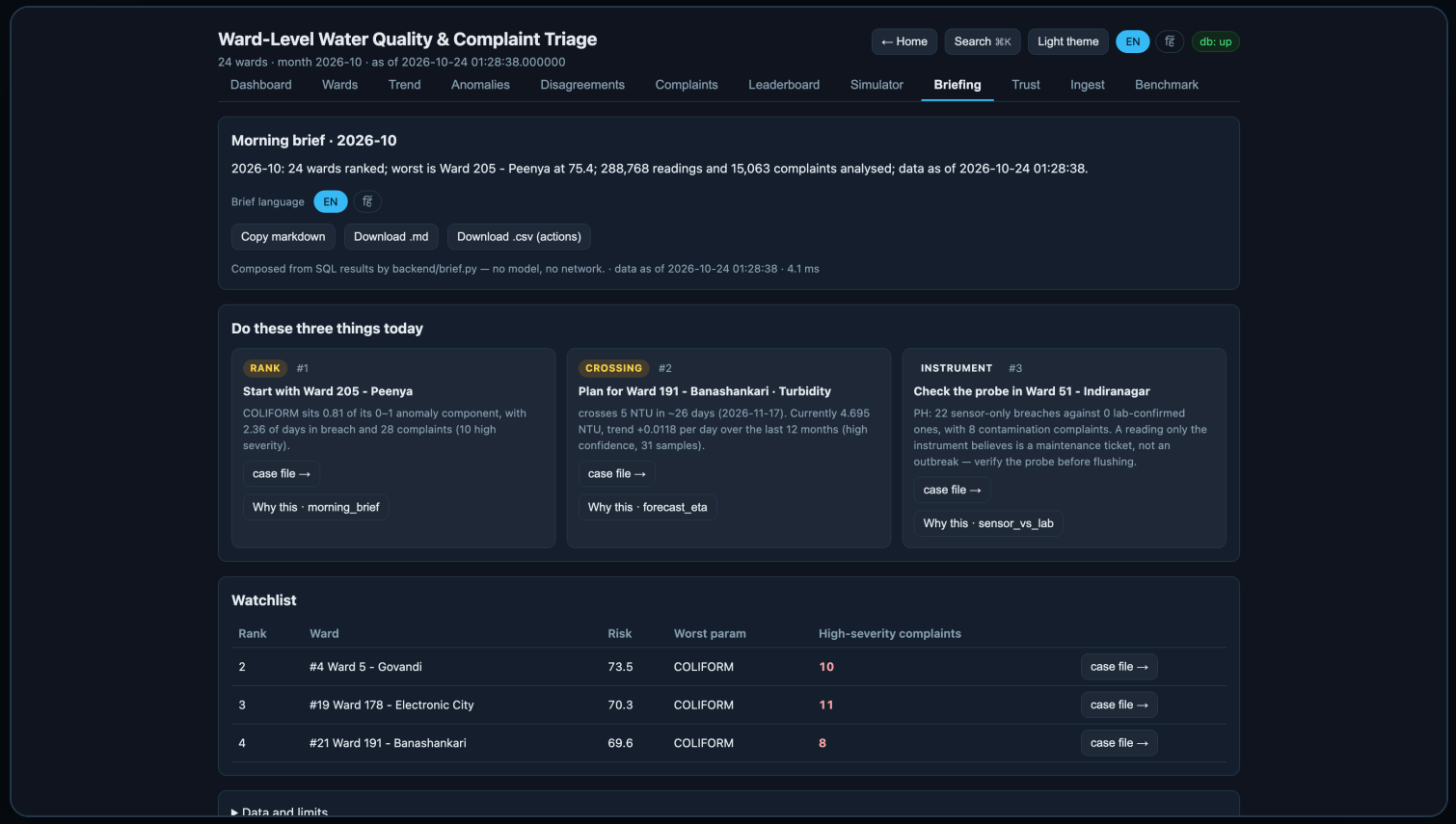
Ward-group exports parsed straight from the raw .txt files · 4 categories · severity independent of category

- **Three weighted rule groups** — contamination, leakage, pressure — plus an explicit `other` fallback so a mis-tag is visible, not hidden.
- **Severity is decided separately** from category: a leak that has run for three days outranks a faint smell.
- **Evidence per message**: the matched phrase is stored in a column you can query — the answer to "why is this tagged contamination" is a row, not a latent vector.
- **No download, no key, no bill**: pure standard library on a 350 KB/s link.

RULE VERSION 2026.10

The rule table is data, so it can be tuned live in front of an audience without touching control flow — and tuned rules re-score all 15,063 messages in one pass.

The brief leaves the app and walks into the officer's day.



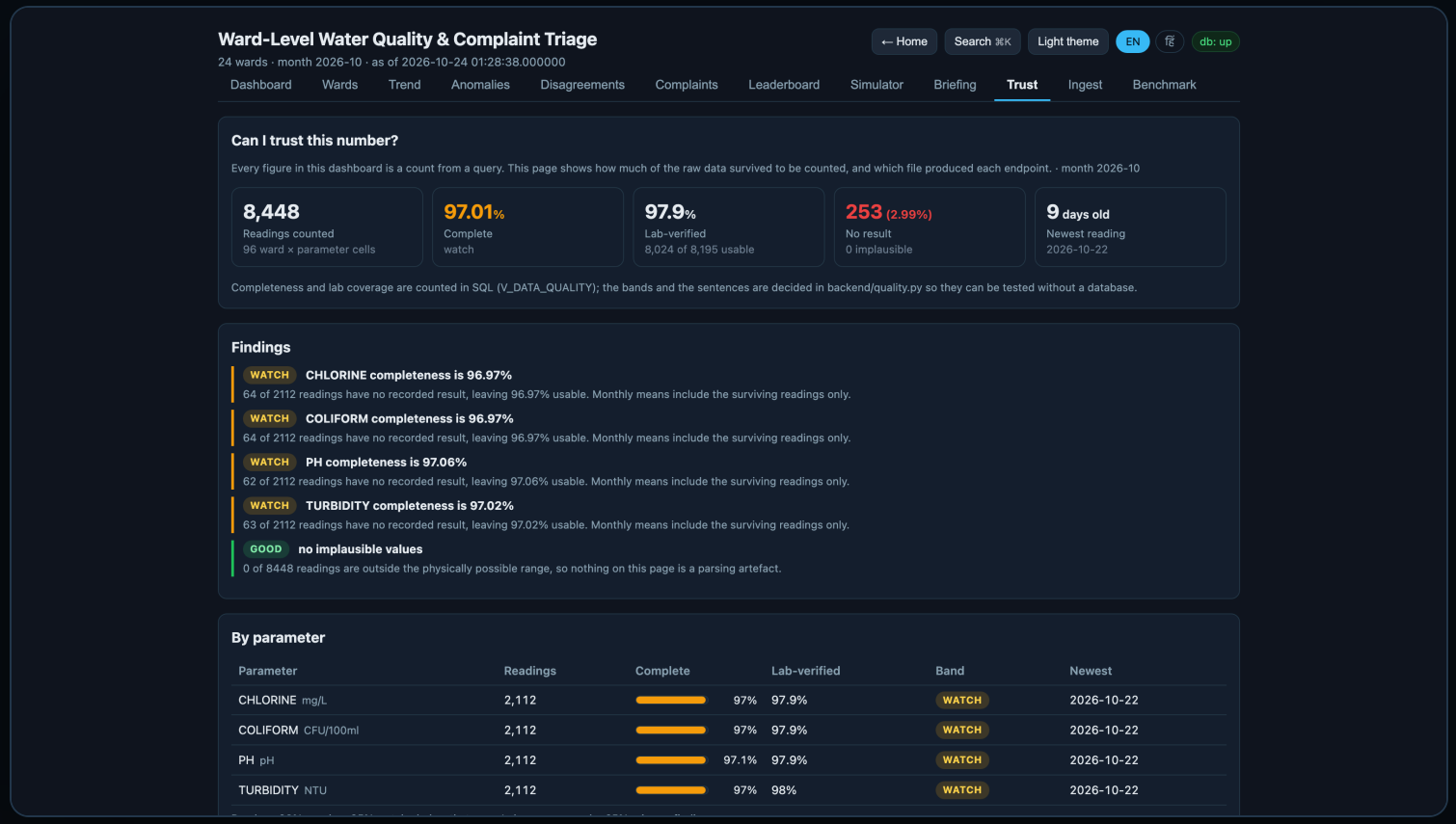
Composed by backend/brief.py from SQL results — no model, no network · 4.1 ms

THREE ACTIONS, RANKED, EACH WITH A REASON

#1 RANK start with the worst ward · **#2 CROSSING** plan for the ward that breaches in 26 days · **#3 INSTRUMENT** check the probe that only the instrument believes.

- **Download** `.md` and an actions `.csv`; copy to clipboard in one click.
- **Calendar**: `/api/calendar.ics` puts every 90-day breach ETA in the officer's calendar with a 7-day alarm.
- **Atom feed**: pull, not push — no SMS bill, no vendor, no account.
- **Hindi brief**, because the officer who reads it at 8 am is not reading an English dashboard.

The dashboard publishes its own defect rate.



V_DATA_QUALITY · completeness and lab coverage counted in SQL, bands decided in testable Python

- **97.01% complete**, 97.9% lab-verified — and the 253 readings with no result are named, counted and left in the denominator.
- **Zero implausible values** of 8,448: nothing on screen is a parsing artefact.
- **Lineage registry**: every figure names the file behind it, opens that SQL in place, and copies the command that reproduces it.
- **9 days old** is shown as a fact, not hidden — freshness is a number, not a slogan.

THE DEMO LINE

"Three percent of our readings are unusable. Here is the number, the cause, and the query — rather than a promise that the data is clean."

UNDER THE HOOD

Every analytic runs inside Exasol.

The application never does the maths — it reads answers the engine has already computed, which is why the whole dashboard loads in under a second on a two-core VM.

SOURCES

What a ULB already has

- Probe / lab CSV logs
- WhatsApp ward exports (.txt)
- Ward master (population, connections)

INGEST

Stdlib parsers, audited

- Tolerant CSV + messy-edge-case parsers
- 169-pattern rule classifier
- LOAD_RUN audit row per load

EXASOL PERSONAL · WARDQ

Where the maths lives

- 10 tables · 10 analytical views
- Lua UDF RISK_SCORE
- 2 materialisations (with / without UDF)
- Window functions: z-score, breach islands, slope

SERVING

22 endpoints, one origin

- FastAPI, stdlib only
- Pre-aggregated reads only
- Same origin, same port — CORS deleted

SURFACE

12-tab PWA + outputs

- Offline service worker (101 lines)
- Hindi chrome · light theme
- .md · .csv · .ics · Atom

the engine does the work · 6.3 GB total RAM · 2.2 GB on disk · no cloud, no account, no API key

10	10	1	11	22	0
TABLES	ANALYTICAL VIEWS	LUA UDF, RISK_SCORE	SQL RECIPES BEHIND "WHY THIS"	JSON ENDPOINTS	THIRD-PARTY RUNTIME DEPS

MEASURED, NOT ASSERTED

Why Exasol: 6.4× faster on the same 288k-row scan.



Scenario	Engine	Rows touched	Elapsed ms	Bytes sent
S0_mirror_build	sqlite	288,768	671	20,041,728
S1_scan_aggregate	exasol	288,768	15	11,231
S1_scan_aggregate	sqlite	288,768	96	4,415
S2_ward_zscore	exasol	752	7	86,359
S2_ward_zscore	python	752	19	39,218

Local Personal VM: 2 vCPU / 6 GB, one data node. No distribution clause is in the DDL at all (a single node makes it meaningless and this Personal build does not parse it); the wins reported are columnar scan, pushdown, pre-aggregation and the result cache.

Exasol Personal · WARDQ schema · rule engine · built for ExaThon India 2026.

Benchmark tab · in-app, re-runnable · 2 vCPU / 6 GB, one data node

- **15 ms vs 96 ms** — monthly aggregate over 288,768 rows: Exasol against the same query in SQLite.
- **7 ms vs 19 ms** — ward z-scores, in-database against the same arithmetic in Python.
- **11,231 bytes** on the wire instead of 20 MB moved to a mirror.
- **671 ms** just to build that mirror — the cost the dashboard never pays.

WHAT WE DO NOT CLAIM

Single node, and no `DISTRIBUTED BY` in the DDL — this Personal build does not parse it, so a distribution story would be fiction. The wins reported are columnar scan, predicate pushdown, pre-aggregation and the result cache.

ENGINEERING RIGOR

Built like it has to survive Monday — 213 tests.

Every claim in this deck has a test behind it, and the database-backed tests run against the live Exasol instance rather than a fixture — the suite cannot go green while the data is broken.

DATA & SQL · 68 TESTS

- **47 SQL snapshot tests** execute against the live database and assert the planted signals still appear.
- **11 row-type** and **10 data-quality** tests guard the Exasol-to-API contract — and DB-backed tests fail loudly, never silently skip.

PARSING & TRIAGE · 64 TESTS

- **24 parser goldens** for the messy WhatsApp edge cases — emoji, Hinglish, forwarded chains.
- **24 ingest** endpoint tests, so a dropped export cannot poison a load.
- **16 rule-table tests**: severities, tie-breaks and the explicit `other` fallback.

OUTPUTS & MEASUREMENT · 81 TESTS

- **15 brief · 13 forecast · 11 calendar** tests cover the numbers the officer actually acts on.
- **8 Atom · 5 lineage** tests keep the feed and the receipts honest.
- **15 benchmark · 14 load** tests keep every figure in this deck reproducible.

REACH & RESTRAINT

- **Installable PWA** with an offline cache — pull the Wi-Fi dongle and it still opens, flagged stale.
- **Hindi chrome and brief**, light theme for a projector-heavy room, CSV export on six tables.
- **Zero new dependencies**: stdlib Python, hand-written JS, 101-line service worker, no router.

SAID OUT LOUD

The limits, before you ask.

Every number in this deck was measured on this machine. The three things that were not are below — and each one is a scope boundary, not a surprise.

The readings are generated

288,768 samples and 15,063 complaints are synthesised in pure SQL, with the monsoon intrusions and probe faults deliberately planted so the detectors have something true to find. The pipeline, the SQL and the detectors are real; the water is not.

One node, no replication

Exasol Personal on 2 vCPU / 6 GB. Fast enough to prove the columnar case; not a municipal HA story. Nothing here is benchmarked against a distributed cluster, and we say so on the benchmark page.

The forecast is a straight line, on purpose

OLS slope over a trailing 12 months with a widening 90-day band — no hydrology model, no rainfall covariates. It is a triage instrument that will be wrong in interesting ways, which is why the band and the confidence label ship with the number.

3% of readings are unusable

253 of 8,448 cells have no result. They are excluded from means, counted in completeness, and displayed on the Trust tab. A detector that pretends to 100% coverage is lying about the network.

WHAT A PILOT WOULD NEED FROM A REAL ULB

Ninety days of probe and lab exports for one ward, the ward population table, and the WhatsApp group the complaints already arrive in. No integration project, no procurement cycle — the ingest path is a folder and a column map.

THE ASK

Give us one ward, ninety days, and one officer's morning.

The dashboard is already built, tested and offline-capable. The next 30 days are about turning a demo dataset into a decision someone actually makes.

DAYS 1–7
Ingest their exports; map their ward and parameter master.

DAYS 8–14
Calibrate thresholds and the four weights with the officer, not for him.

DAYS 15–21
Hindi brief, calendar alarms, CSV hand-off into their existing process.

DAYS 22–30
Publish a baseline: which wards move, and whether anyone acted early.

WARDQ · ward-level water quality & complaint triage

ExaThon India 2026 · Civic / Sustainability

₹0

Software cost of the pilot. Exasol Personal, stdlib Python and hand-written JS — no licence, no cloud bill, no per-ward fee.

LIVE
`localhost:8000`

PROOF
24 wards · 288,768 readings · 15,063 complaints · 213 tests

REPEATABLE
Every figure on screen opens the SQL behind it

THANK YOU · QUESTIONS WELCOME, INCLUDING THE HOSTILE ONES